

STA304 Midterm Test (Lec0101)

Tuesday October 17, 2023

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Instructions:

- Total points: 20
- The test is 75 minutes long.
- There are 13 pages including this page.
- Questions begin on the next page.
- Only this exam book will be marked. Write your answers only in the space provided in this booklet. Any work on the Formula Sheet booklet will not be graded.
- Total questions: 13 (10 multiple choice, 3 short answer)
- Please ensure you **fill in your multiple Choice selections (Q1 - Q10) in the scantron on page 13**. Multiple choice answers NOT on the scantron will NOT be graded.
- Please use a calculator and do not round numbers in calculations until the very end, to avoid rounding error.
- In the multiple choice options, please select the closest answer (if the options are numeric) or the most appropriate answer (if the options are text).
- If you believe there to be a typo please still answer the question (see bullet point above), but you can list the question here and we will look into it after:

Multiple Choice Questions

Q1 (1 point)

A company of 15 employees wishes to take a sample of 5 of the employees. Read the following three statements:

Statement 1: If they take a systematic sample then the probability of selecting any particular sample is $\frac{1}{5}$.

Statement 2: If they take a simple random sample without replacement then the probability of selecting any particular sample is $\frac{1}{3003}$.

Statement 3: If they take a simple random sample with replacement then the probability of selecting any particular sample is $\frac{1}{759375}$.

Select ONE of the options below, to best describe which of the above statements are TRUE.

- a. Only Statement 2 is correct.
- b. Only Statements 1 and 2 are correct.
- c. Only Statements 2 and 3 are correct.
- d. All three statements are correct.
- e. None of the three statements are correct.

Q2 (1 point)

Read the following three statements:

Statement 1: Snowball sampling is a non-probability-based sampling technique which is useful for hidden populations.

Statement 2: If a high school population has 200 male students and 200 female students, then handing out a survey to the first 50 male students and first 50 female students who enter the school that day is an example of stratified random sampling.

Statement 3: Cluster sampling is a probability-based sampling technique.

Select ONE of the options below, to best describe which of the above statements are TRUE.

- a. Only Statement 1 is correct.
- b. Only Statements 1 and 3 are correct.
- c. Only Statements 2 and 3 are correct.
- d. All three statements are correct.
- e. None of the three statements are correct.

Q3 (1 point)

A hospital is planning to conduct a Systematic Random Sample on patients in a particular department over a week. They wish to estimate the *total* times all patients spend in the department over a week. They would like the total to be correct within 14 hours. Last week there were 82 patients in the department, and the standard deviation of the time spent in the department of each patient was 1.92 hours. Assume you want to ensure to calculate the total with at least 90% confidence, what sample size should be taken?

- a. 10
- b. 40
- c. 57
- d. 67
- e. 70

Q4 (1 point)

Which ONE of the following are FALSE?

- a. When using cluster sampling we assume that clusters are similar/homogeneous.
- b. When using stratified sampling we are sampling in such a way that we will have representation from each strata in our sample.
- c. An example of a linear regression model is $y = \beta_0 + \beta_1 x + \epsilon$
- d. An example of a logistic regression model is $\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x$.
- e. A Frequentist framework uses a prior distribution to make inference on the parameter of interest, and a Bayesian framework does not use any priors.

Q5 (1 point)

The professor of the class surveyed the students in STA304 and stratified by lecture section. The class has $N = 500$ students in total, 230 of which are in the Morning section. The professor is interested in estimating the mean hours slept of students the day before the assignment due date. Use the following summary table to calculate the *lower bound* of the 95% confidence interval for the population mean hours slept prior to the assignment due date.

Lecture section	n	mean	sd
Morning	23	5.61	1.68
Afternoon	27	7.02	1.90

- a. 5.8975
- b. 6.2568
- c. 6.2752
- d. 6.3079
- e. 6.8453

Q6 (1 point)

Which ONE of the following is an example of a double-barreled question?

- a. "How often do you attend lecture and which section do you attend?"
- b. "Why is this course so good?"
- c. "What is your age?"
- d. "Are you currently enrolled in STA304?"
- e. All of the above are double-barreled questions.

Data for Questions 7, 8, 9, and 10

Recall that the `UCLdata` from lecture contained the following variables:

- `admit`: (binary) 1 indicates that the student was admitted to graduate school, and 0 indicates they were not admitted.
- `gre`: (numeric) the students' recorded GRE score.
- `gpa`: (numeric) the students' recorded GPA score.
- `rank`: (categorical) Indicates whether the student attended a Rank 1, Rank 2, Rank 3, or Rank 4 university for their undergraduate studies.

Recall, this is what the data looked like:

```
glimpse(UCLdata)
```

```
## Rows: 400
## Columns: 4
## $ admit <int> 0, 1, 1, 1, 0, 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 0, 0, 0, 0, 1, 0, 1~
## $ gre    <int> 380, 660, 800, 640, 520, 760, 560, 400, 540, 700, 800, 440, 760, ~
## $ gpa    <dbl> 3.61, 3.67, 4.00, 3.19, 2.93, 3.00, 2.98, 3.08, 3.39, 3.92, 4.00~
## $ rank   <int> 3, 3, 1, 4, 4, 2, 1, 2, 3, 2, 4, 1, 1, 2, 1, 3, 4, 3, 2, 1, 3, 2~
```

```
head(UCLdata)
```

```
##   admit gre  gpa rank
## 1     0 380 3.61    3
## 2     1 660 3.67    3
## 3     1 800 4.00    1
## 4     1 640 3.19    4
## 5     0 520 2.93    4
## 6     1 760 3.00    2
```

Q7 (1 point)

If we wish to compare the sample size, admittance rate, as well as the mean GRE score, and variance of the GRE score of applicants from Rank 1 and Rank 2 universities, which code chunk would be most appropriate?

a. The code chunk below:

```
UCLdata %>%
  filter(rank==1 & rank==2) %>%
  group_by(rank) %>%
  mutate(n=n(), a=mean(admit), y=mean(gre), v=var(gre))
```

b. The code chunk below:

```
UCLdata %>%
  select(rank==1 & rank==2) %>%
  filter(rank) %>%
  summarize(n=n(), a=mean(admit), y=mean(gre), v=var(gre))
```

c. The code chunk below:

```
UCLdata %>%
  filter(!rank==3 | !rank==4) %>%
  group_by(rank) %>%
  summarize(n, mean(gre), var(gre))
```

d. The code chunk below:

```
UCLdata %>%
  select(rank, gre) %>%
  filter(rank==1 | rank==2) %>%
  group_by(rank) %>%
  summarize(n=n(), a=mean(admit), y=mean(gre), v=var(gre))
```

e. The code chunk below:

```
UCLdata %>%
  filter(rank==1 | rank==2) %>%
  group_by(rank) %>%
  summarize(n=n(), a=mean(admit), y=mean(gre), v=var(gre))
```

Q8 (1 point)

Assume that in 2023, there were $N=8078$ undergraduate applications made to UCLA graduate programs. We wish to make inference on the ratio of GRE scores and the GPA scores of students. The following code was run to perform this analysis and the output is below.

```
N=8078
n=nrow(UCLAdata)

my.design <- svydesign(id=~1,
                      data=UCLAdata,
                      fpc=rep(N, n))

svyratio(~gre, ~gpa, design=my.design)

## Ratio estimator: svyratio.survey.design2(~gre, ~gpa, design = my.design)
## Ratios=
##      gpa
## gre 173.3679
## SEs=
##      gpa
## gre 1.564685
```

Which ONE of the following statements about the above scenario (code and output) is TRUE?

- a. In this example we are assuming that a stratified random sampling in the approach was used, based on the input in `svydesign()`.
- b. In this example we can see that GPA score is estimated to be 173 times larger than the GRE score.
- c. We would need to test $H_0 : R = 0$ vs $H_A : R \neq 0$, to assess if GRE and GPA scores are significantly different from one another.
- d. We would need to test $H_0 : R = 1$ vs $H_A : R \neq 1$, to assess if GRE and GPA scores are significantly different from one another.
- e. The sample size is 8078.

Q9 (1 point)

The following code was used to estimate the GRE score of students based on their GPA score and the rank of their university.

```
UCLA_gre_model <- lm(gre ~ gpa + as.factor(rank), data=UCLAdata)
summary(UCLA_gre_model)$coefficients
```

##	Estimate	Std. Error	t value	Pr(> t)
## (Intercept)	206.986270	50.50776	4.0981081	5.057237e-05
## gpa	117.232423	14.08652	8.3223102	1.421978e-15
## as.factor(rank)2	-5.054814	16.16565	-0.3126885	7.546823e-01
## as.factor(rank)3	-34.556549	16.68155	-2.0715427	3.895676e-02
## as.factor(rank)4	-25.856189	18.89300	-1.3685591	1.719150e-01

Which ONE of the following statements are TRUE?

- a. The value -5.055 means that for every one unit increase in rank, we expect a 5.055 unit decrease in the GRE scores (for students attending universities of the same rank).
- b. We cannot calculate the predicted GRE score of a student at a Rank 1 university with a 3.5 GPA.
- c. We would expect that students at a rank 4 university have a GRE score of -25.856 on average.
- d. The model we are estimating is $y = \beta_1 x_{gpa} + \beta_2 x_{rank1} + \beta_3 x_{rank2} + \beta_4 x_{rank3} + \beta_5 x_{rank4}$.
- e. We would expect that students at a rank 1 university have a GRE score of about 34.557 points higher than students at a rank 3 university, on average.

Q10 (1 point)

The code below was used to estimate the admittance rate of students, into graduate school, based on their GRE score and the rank of their undergraduate university.

```
UCLA_admit_model <- glm(admit ~ gre + as.factor(rank),  
                        data=UCLAdata, family="binomial")
```

```
summary(UCLA_admit_model)$coefficients
```

```
##              Estimate Std. Error  z value    Pr(>|z|)  
## (Intercept)   -1.802364503 0.672982151 -2.678176 0.0074024357  
## gre           0.003223909 0.001019285  3.162913 0.0015619926  
## as.factor(rank)2 -0.721736747 0.313033472 -2.305622 0.0211317831  
## as.factor(rank)3 -1.291305200 0.340774710 -3.789322 0.0001510588  
## as.factor(rank)4 -1.602053861 0.414932115 -3.861002 0.0001129228
```

Based on the output above, calculate the predicted probability of admittance for each of the two students listed below, and select the ONE statement below that is TRUE.

Student	GRE	rank
Sam	603	2
Emily	791	1

- a. Sam's predicted probability of admittance is 0.5801.
- b. Emily's predicted probability of admittance is 0.7477.
- c. Sam's predicted probability of admittance is 0.6411.
- d. Emily's predicted probability of admittance is 0.6787.
- e. We cannot calculate the predicted probability of admittance for either student.

Short Answer Question

Q11 (3 points total)

A local high school is interested in conducting a simple random sample of its student population, $N = 460$. They have a spreadsheet of the population and they also have each column/variable stored as its own vector. The first six rows of the population data look like this:

```
## # A tibble: 6 x 3
##   Name          StudentID Grade
##   <chr>         <chr>    <dbl>
## 1 Mary Adams    adams12     10
## 2 Mary Adams    adams13      9
## 3 Blue Bokowski bokowsk     10
## 4 Katy Calee    calee456     11
## 5 Cody Cenderson cenderson1    9
## 6 Maya Culph    culph1      12
```

Write a line of code to select a simple random sample, with replacement, of 30 students.

Short Answer Question

Q12 (3 points total)

You are working on analyzing some survey data collected on patients with a rare disease ($N=50$, and $n=15$). You are contemplating whether or not you should publish your summary statistics (i.e., histograms, minimums, means, etc) on a public website in hopes of ensuring your work is reproducible. Identify one right and one impact to consider in this scenario and in one or two sentences comment on the appropriateness of whether or not the study can be made fully reproducible.

Impact:

Right:

Comments:

Short Answer Question

Q13 (4 points total)

Assume you have a simple random sample without replacement of $Y_1, Y_2, \dots, Y_n$ taken from the population of $\{u_1, u_2, \dots, u_N\}$. Recall from class:

$$E(Y_i) = \mu \qquad \text{Var}(Y_i) = \sigma^2$$

$$E(\bar{Y}) = \mu \qquad \text{Var}(\bar{Y}) = \frac{\sigma^2(N-n)}{n(N-1)}$$

Derive $E(n\bar{Y}^2)$. Show your work. Write your final answer as an expression of μ , σ , N and n .